E4.2.1 Pressure gain in the middle ear

Sound reaches the eardrum through the pinna and ear canal, making it vibrate. The ossicles transmit vibration to the oval window and inner-ear fluid. Their lever action increases force, while the oval window's smaller area raises pressure further.

The Eustachian tube connects the middle ear to the throat. When open, it helps equalise pressure across the eardrum. During aircraft ascent or descent, outside pressure may change before middle-ear pressure adjusts. The pressure difference can cause pain and restrict vibration. Swallowing helps open the tube.



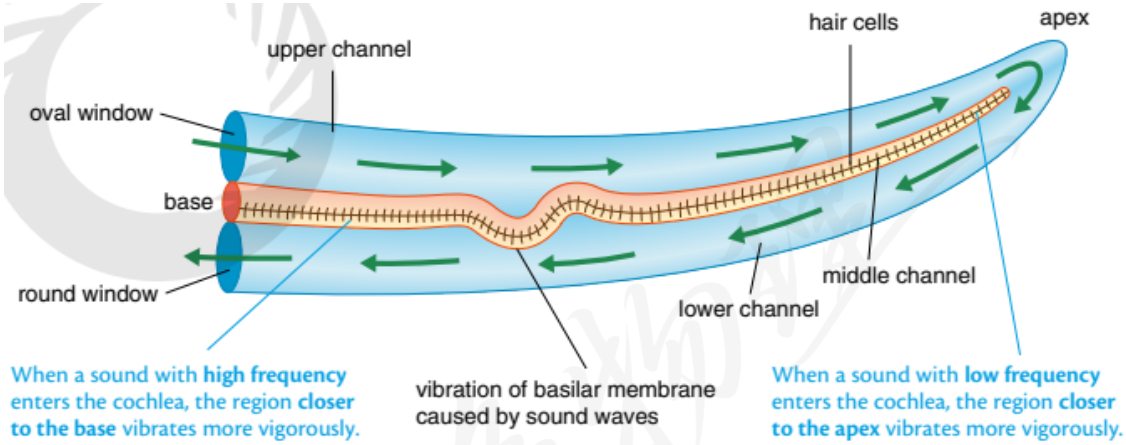
Fig 2.1d *Combined effect of the lever action and the area ratio of the eardrum to the oval window.*

Lever action increases force and the smaller oval-window area further increases pressure. The illustrated factors multiply to give the pressure gain.

E4.2.2 Frequency analysis in the cochlea

Oval-window vibration travels through cochlear fluid and vibrates the basilar membrane. Bending stimulates hair cells, which convert mechanical vibration into electrical signals sent to the brain through the auditory nerve.

High frequencies produce greatest vibration near the base and oval window; low frequencies peak nearer the apex. The brain uses the location of stimulated hair cells to identify pitch. Vibration amplitude and neural signals also affect loudness.



The cochlear base responds most strongly to higher frequencies and the apex to lower frequencies.

E4.2.3 Sound intensity and distance

Sound intensity is energy crossing unit area perpendicular to propagation per second, measured in W m^{-2}. For an isotropic point source in a free field with negligible absorption, energy spreads over spheres whose area grows with distance squared. Doubling distance quarters intensity. Reflection and absorption can alter this relationship in real environments.

$$I = \frac{E}{At} = \frac{P}{A}$$

$$I = \frac{P}{4\pi r^2}$$

I: average sound intensity (W m^{-2}); E: sound energy (J); A: perpendicular area (m^2); t: time (s); P: sound power (W).

Isotropic point source in a free field, negligible absorption. I: W m^{-2}; acoustic power P: W; distance r: m.

E4.2.4 Sound intensity level and decibels

Sound intensity level compares intensity with a reference using a base-ten logarithm. Multiplying intensity by ten adds 10 dB, not ten times the decibel value. Zero dB means equal to the reference intensity, not silence. For independent incoherent sources, add intensities before converting to decibels.

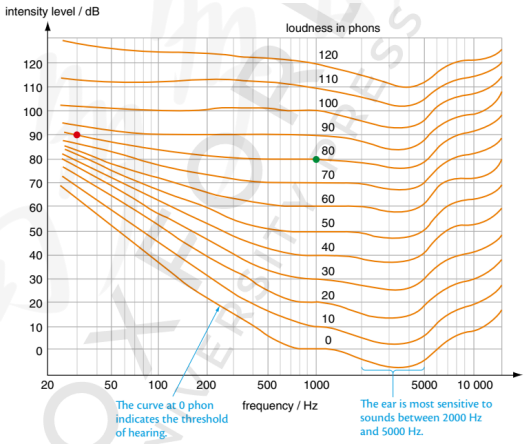
$$L = 10\log_{10}\left(\frac{I}{I_0}\right)$$

$$I_0 = 10^{-12}\ \mathrm{W\,m^{-2}}$$

Sound intensity level L (dB); I and I_0 in identical units. Base-10 logarithm; the ratio inside the logarithm is dimensionless.

E4.2.5 Equal-loudness curves

Points on one equal-loudness curve sound equally loud even though different frequencies require different decibel levels. At the specified frequency and level, identify the curve passing through the point. If all frequencies increase by the same dB amount, reread each point. This does not automatically give the same perceived loudness increase.



Equal-loudness contours: the same perceived loudness requires different sound levels at different frequencies.

E4.2.6 Hearing threshold and hearing loss

A threshold curve gives the minimum sound level just audible at each frequency. An upward shift means a louder sound is needed at that frequency, indicating reduced sensitivity. Compare frequency by frequency rather than assigning one dB value to all pitches. An increased threshold is not a change in source loudness.

E4.2.7 Piezoelectric transducers

An alternating voltage makes a piezoelectric crystal expand and contract, emitting ultrasound pulses. Returning echoes deform it and produce voltage signals. The same element can alternate between transmission and reception, with time after each pulse to receive echoes.

Crystal thickness is selected for the intended ultrasound frequency so its mechanical vibration suits that frequency. Different operating frequencies therefore require corresponding crystal designs.

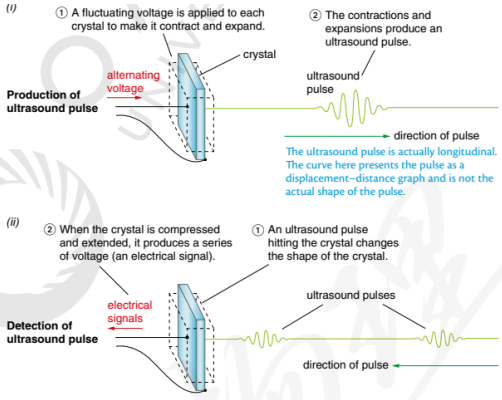


Fig 2.2g (i) Production and (ii) detection of ultrasound pulse by a transducer.

Above: alternating voltage generates a sound pulse. Below: an echo deforms the crystal and generates voltage.

E4.2.8 Acoustic impedance and reflection

Acoustic impedance is density times sound speed. At normal incidence, a greater impedance mismatch generally gives a greater reflected intensity fraction. The reflection coefficient is an intensity ratio, not an amplitude ratio, and lies between zero and one. Coupling gel reduces strong reflection from an air gap between probe and skin.

$$Z = \rho c$$

$$\alpha = \frac{I_r}{I_i}$$

$$\alpha = \frac{I_r}{I_i} = \left(\frac{Z_2 - Z_1}{Z_2 + Z_1}\right)^2$$

Acoustic impedance Z (kg m^{-2} s^{-1}); density ρ (kg m^{-3}); sound speed c in the medium (m s^{-1}). Here c is not light speed.

α: intensity reflection coefficient (dimensionless); I_r: reflected intensity; I_i: incident intensity, called I_0 in the textbook. Both intensities in W m^{-2}.

Normal incidence at a plane boundary between two media; Z_1, Z_2: acoustic impedances in the same units. α is the reflected INTENSITY fraction, not amplitude.

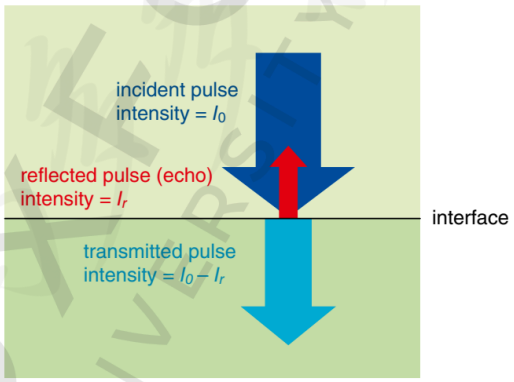


Fig 2.2h *Reflection and transmission of ultrasound pulse.*

Incident intensity divides into reflected and transmitted parts. A larger impedance mismatch gives a larger reflected fraction at normal incidence.

E4.2.9 Distance from ultrasound echoes

An ultrasound pulse travels to an interface and back, so depth equals speed times round-trip time divided by two. The time difference between echoes can give separation of interfaces on the same path. For different materials along the path, calculate each travel time using the appropriate sound speed.

$$\Delta d = \frac{1}{2} c \Delta t$$

Pulse–echo: Δt is the difference in round-trip echo times (s); c: sound speed (m s^{-1}); Δd: separation along the beam (m). Factor ½ accounts for outward and return paths.

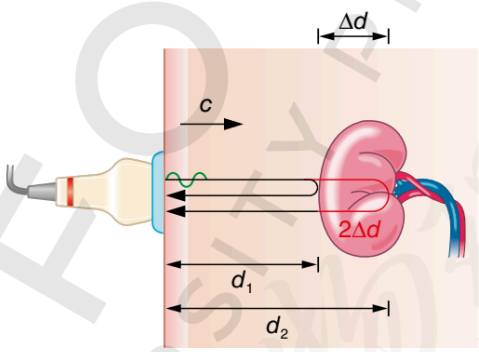


Fig 2.3d *An ultrasound pulse is reflected from the front and back surfaces of a kidney.*

The back-surface echo travels an extra outward and return distance of 2Δd. Therefore the time difference gives Δd = cΔt/2.

E4.2.10 A-scan and B-scan displays

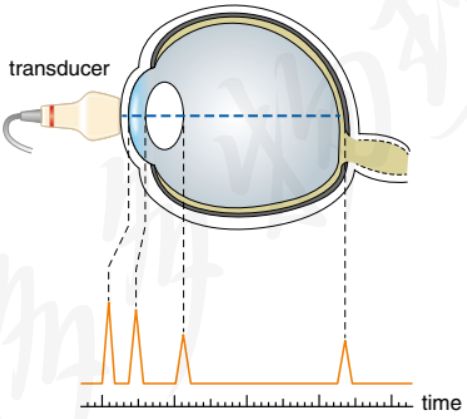
An A-scan displays echo peaks: horizontal position gives arrival time and height gives signal strength. At sound speed c, interface depth is $\frac{ct}{2}$ because the pulse travels out and back. A B-scan converts arrival time to position and strength to brightness, combining scan lines into a two-dimensional section that can update in real time. A dark region means weaker received echoes, not necessarily absent tissue or disease.

Ultrasound frequency and penetration

Within the same tissue, higher-frequency ultrasound generally attenuates faster and has less penetration depth. Lower frequency reaches deeper but usually sacrifices detail resolution. Balance target depth against resolving requirements when choosing frequency.

$$f \uparrow \implies \text{attenuation} \uparrow \implies \text{penetration} \downarrow$$

In the same tissue, higher ultrasound frequency generally increases attenuation and reduces penetration depth.



Echo positions and peak heights in an A-scan represent arrival time and signal strength.

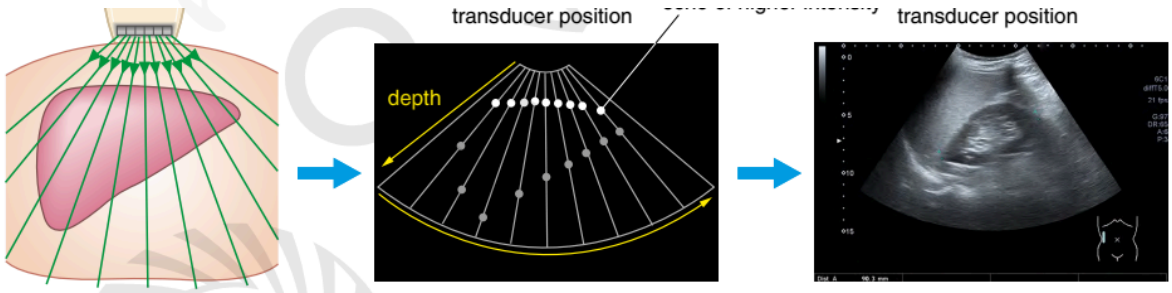


Fig 2.3g Production of a B-scan image.

A B-scan combines multiple echo lines into a two-dimensional section.

E4.2.11 Ultrasound pulses and axial resolution

Nearby interfaces along the beam are distinguishable only if their echoes are sufficiently separated in time. Shorter pulses improve axial resolution. The limiting separation is about half the spatial pulse length. Higher frequency directly shortens a pulse only when its number of cycles is held fixed.

$$\Delta d_{\min} \approx \frac{\text{spatial pulse length}}{2}$$

Shorter ultrasound pulses improve axial resolution. At a fixed number of cycles per pulse, higher frequency gives shorter pulses and a smaller minimum resolvable separation.

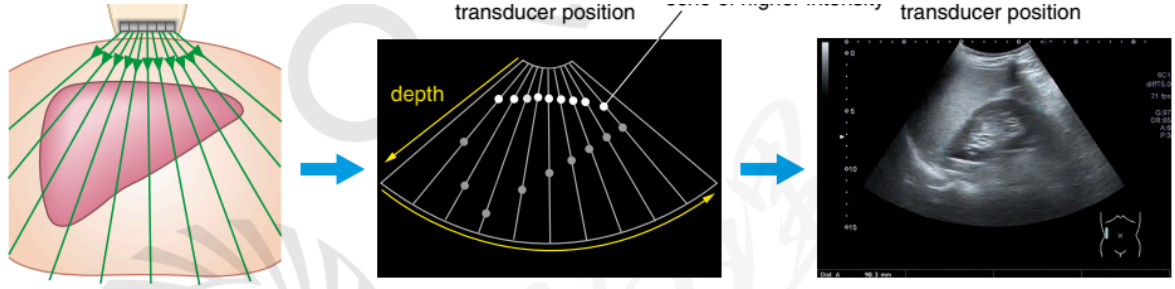


Fig 2.3g Production of a B-scan image.

A B-scan combines multiple echo lines into a two-dimensional section.

E4.2.12 Comparing medical imaging methods

First decide whether the target is an accessible internal surface, a tissue section or real-time movement. Then compare penetration, field of view, invasiveness and ionising radiation. Ultrasound is non-ionising and can show live movement, but passes poorly through bone and gas. Fibre-optic endoscopy directly views accessible cavities and can take samples, but requires insertion and has limited reach and view. Relate each limitation to the specified target and purpose.